

3A Load Switch with TRCB Function

General Description

The ET3158A advanced load management switches target applications requiring a highly integrated solution. It disconnects loads powered from DC Power Rail (<6V) with stringent off-state current targets and high load capacitances (up to 200uF). Each switch consists of slew-rate controlled low-impedance MOSFET Switch and other integrated analog features. The slew-rate controlled turn-on characteristic prevents inrush-current and the resulting excessive voltage droop on power rails.

The ET3158A has True Reverse Current Blocking (TRCB) function blocking unwanted reverse current from OUT to IN during ON/OFF state. These devices have exceptionally low off-state current drain (<1uA max) which facilitate compliance in very low stand-by power applications. The input voltage range operates from 1.5V to 6.0V DC to fulfill a wide range of applications in consumer, optical, medical, storage, portable, and industrial device power management. Switch control is managed by a logic input (Active HIGH) capable of interfacing directly with low voltage control signal/GPIO with no external pull-down resistor required.

ET3158A is offered DFN4(1.2mm ×1.6mm) package, which is ideal for small form factor portable equipment.

Features

- 1.5V to 6 V Operation Voltage Range
- Low Quiescent Current is 8uA Typical
- Slew Rate/Inrush Control with tR is 2.7ms Typical
- Typical RDS(ON) is 75mΩ at V_{IN} = 3.6V
- True Reverse Current Blocking (TRCB)
- ESD Protection Complies with JESD22 Standard
 - HBM: ±4000V Pass (JEDEC JS-001)
 - CDM: ±1500V Pass (JEDEC JS-002)
- Part No. and Package Information

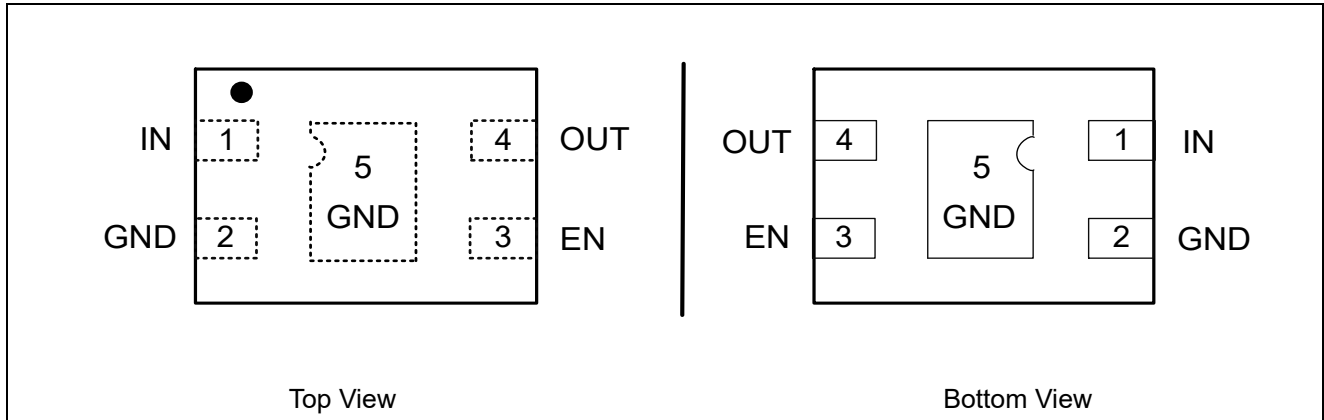
Part No.	Package	Packing Option	MSL
ET3158A	DFN4 (1.2mm ×1.6mm)	Tape and Reel, 3K/Reel	Level 1

Application

- PDAs/Smart Phones
- Notebook/Computers
- Portable Media Players
- Digital Camera
- GPS Navigation Devices
- Data Storage Devices
- Optical, Industrial, Medical, And Health-Care Devices

ET3158A

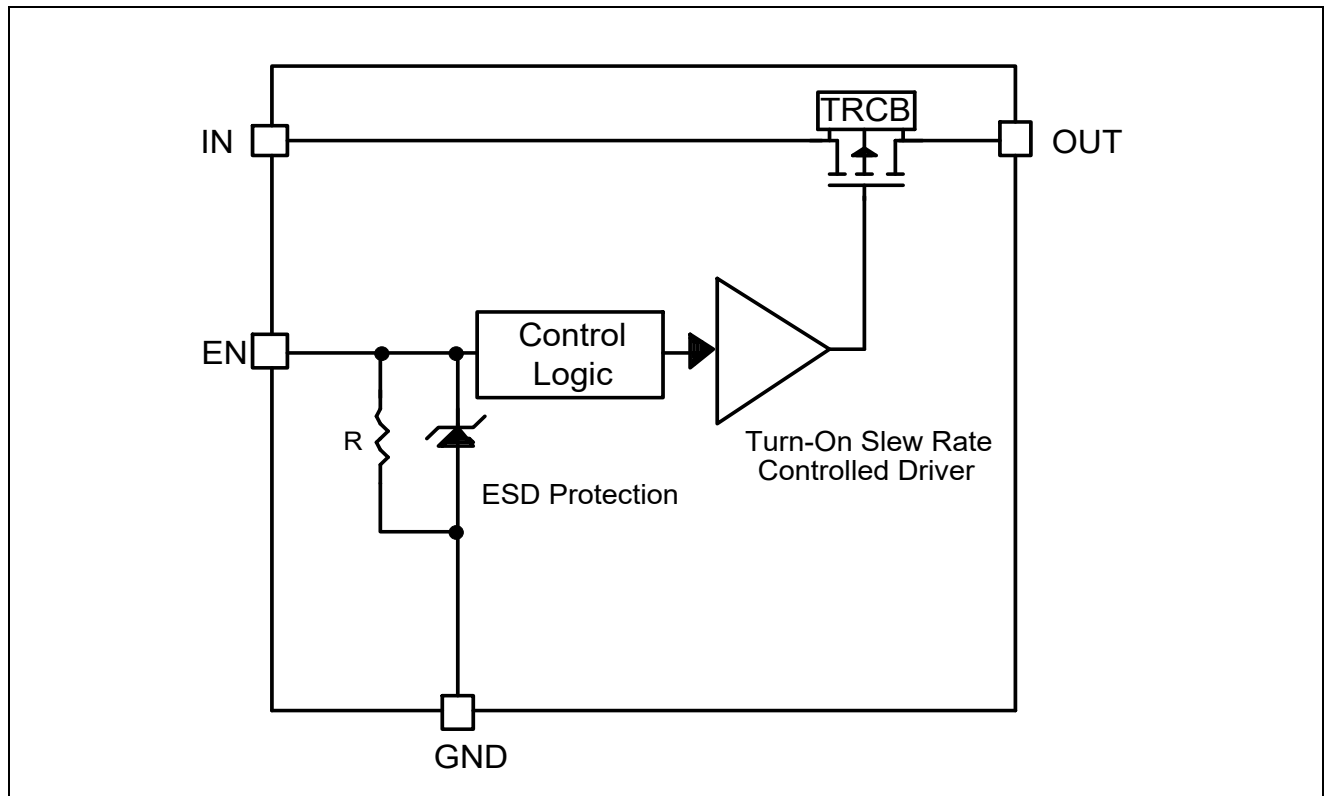
Pin Configuration



Pin Function

Pin Number	Name	Function
1	IN	This is The Input Pin of The Switch
2, 5	GND	Ground Connection
3	EN	Enable Input
4	OUT	This is The Output Pin of The Switch

Block Diagram



ET3158A

Functional Description

The ET3158A is low- R_{ON} P-channel load switches with controlled turn on and TRCB (True Reverse Current Blocking). The core of each device is a P-channel MOSFET and controller capable of functioning over a wide input operating range of 1.5 to 6.0V. The EN pin, an active HIGH GLOP input, controls the state of the switch. TRCB functionality blocks unwanted reverse current during ON and OFF when OUT higher than IN is applied.

Input Capacitor

To limit the voltage drop on the input supply caused by transient inrush current when the switch turns on into a discharged load capacitor or short-circuit, a capacitor must be placed between the IN and GND pins. At least 1 μ F ceramic capacitor, C_{IN} , placed close to the pins is usually sufficient. Higher-value C_{IN} can be used to reduce the voltage drop in higher-current applications.

Output Capacitor

At least 0.1 μ F capacitor, C_{OUT} , should be placed between the OUT and GND pins. This capacitor prevents parasitic board inductance from forcing OUT below GND when the switch is on.

Board Layout

For best performance, all traces should be as short as possible. To be most effective, the input and output capacitors should be placed close to the device to minimize the effect that parasitic trace inductance may have on normal and short-circuit operation.

Using wide traces or large copper planes for all pins (IN, OUT, EN, and GND) helps minimize the parasitic electrical effects along with minimizing the case ambient thermal impedance.

Pulse Current Capability

The device is mounted on the evaluation board shown in the PCB layout section. It is loaded with pulses of 6 A and 1ms for periods of 4.6ms.

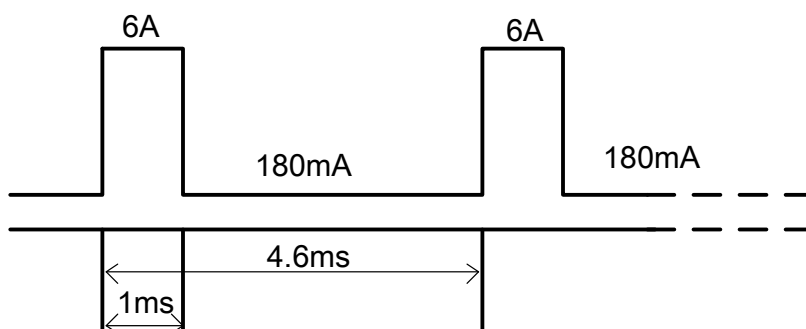


Figure1. Pulse Current Diagrammatic Drawing

The ET3158A can safely support 6A pulse current repetitively at 25 °C.

Switch Non-Repetitive Pulsed Current

The ET3158A can withstand inrush current of up to 15A for 100 μ s at 25 °C when heavy capacitive loads are connected and the part is already enabled.

ET3158A

Absolute Maximum Ratings

Parameter		Value	Unit
Supply Input Voltage (V _{IN}) ⁽¹⁾		- 2 to 7	V
Enable Input Voltage (V _{EN}), Input Resistance is Greater Than 1kΩ		- 2 to 7	
Maximum Continuous Switch Current (I _{MAX}) ⁽³⁾		3	A
Maximum Repetitive Pulsed Current (1ms, 10% Duty Cycle) ⁽³⁾		6	
Maximum Non-Repetitive Pulsed Current (100μs, V _{EN} = Active) ⁽³⁾		15	
ESD/Electrostatic Discharge Capability	Human Body Model, JEDEC JS-001	4.0	KV
	Charged Device Model, JEDEC JS-002	1.5	
	ESD Withstand Voltage IEC61000-4-2	8.0(Contact)	
Junction Temperature (T _J)		- 40 to 150	°C
Thermal Resistance (θ _{JA}) ⁽²⁾		170	°C/W
Power Dissipation (P _D)		735	mW

Notes:

- ET3158A can pass the 10V test (Instant Contact): can support up to 100ms 10V pulse.
- Device mounted with all leads and power pad soldered or welded to PC board.
- T_A = 25 °C.

Recommended Operating Range

Parameter	Value	Unit
Input Voltage Range (V _{IN})	1.5 to 6.0	V
Operating Junction Temperature Range (T _A)	-40 to 85	°C

Electrical Characteristics

(V_{IN} = 5V, T_A = -40°C to 85°C, Typical values are at T_A = 25°C)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
V _{IN}	Operating Voltage		1.5		6	V
I _{Q_ON}	Quiescent Current	V _{IN} = 1.8V, V _{EN} = Active		4		uA
		V _{IN} = 2.5V, V _{EN} = Active		6		
		V _{IN} = 3.6V, V _{EN} = Active		8		
		V _{IN} = 4.3V, V _{EN} = Active		10		
		V _{IN} = 5.0V, V _{EN} = Active		11	14	
I _{Q_OFF}	Off Supply Current	V _{EN} = Inactive, OUT = Open			1	
I _{DS_OFF}	Off Switch Current	V _{EN} = Inactive, OUT = GND			1.2	

ET3158A

Electrical Characteristics(Continued)

($V_{IN} = 5V$, $T_A = -40^{\circ}C$ to $85^{\circ}C$, Typical values are at $T_A = 25^{\circ}C$)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
$R_{DS(on)}$	On-Resistance $T_A = 25^{\circ}C^{(a)}$	$V_{IN} = 1.8V$, $I_L = 100mA$		115		mΩ
		$V_{IN} = 2.5V$, $I_L = 100mA$		89		
		$V_{IN} = 3.6V$, $I_L = 100mA$		75		
		$V_{IN} = 4.3V$, $I_L = 100mA$		70		
		$V_{IN} = 5.0V$, $I_L = 100mA$		67		
V_{IL}	EN Input Low Voltage	$V_{IN} = 1.8V$ to $5.5V$			0.6	V
		$V_{IN} = 2.6V$ to $5.5V$			0.8	
V_{IH}	EN Input High Voltage	$V_{IN} = 1.8V$ to $5.5V$	1.5			
I_{SINK}	EN Input Leakage	$V_{EN} = 5.5V$, $V_{IN} = 0V$	-10		10	uA
		$V_{EN} = 0V$, $V_{IN} = 5.5V$	-1		1	
R_{ON_PD}	Pull-Down Resistance at EN pin	$V_{IN} = 1.5V$ to $6.0V$, $T_A = -40$ to $85^{\circ}C$		1		MΩ
V_{T_RCB}	RCB Protection Trip Point	$V_{OUT} - V_{IN}$		45		mV
V_{R_RCB}	RCB Protection Release Trip Point	$V_{IN} - V_{OUT}$		25		mV
	RCB Hysteresis			70		mV
I_{SD_OUT}	V_{OUT} Shutdown Current	$V_{EN} = 0V$, $V_{OUT} = 5.5V$, $V_{IN} = \text{Short to GND}$			2	μA
T_{RCB_ON}	RCB Response Time when Device EN	$V_{OUT} - V_{IN} = 100mV$ $V_{EN} = \text{High}$		4.0		μs
T_{RCB_OFF}	RCB Response Time Device OFF	$V_{OUT} - V_{IN} = 100mV$ $V_{EN} = \text{Low}$		2.5		μs
t_{DON}	Turn-On Delay ^(a,b)	$V_{IN} = 4.5V$, $R_L = 5\Omega$, $C_L = 100\mu F$, $T_A = 25^{\circ}C$		1		ms
t_R	V_{OUT} Rise Time ^(a,b)			2		ms
t_{ON}	Turn-On Time ^(a,c)			3		ms
t_{DON}	Turn-On Delay ^(a,b)	$V_{IN} = 4.5V$, $R_L = 150\Omega$, $C_L = 100\mu F$, $T_A = 25^{\circ}C$		1		ms
t_R	V_{OUT} Rise Time ^(a,b)			1.5		ms
t_{ON}	Turn-On Time ^(a,c)			2.5		ms
t_{DOFF}	Turn-Off Delay ^(a,b)	$V_{IN} = 4.5V$, $R_L = 150\Omega$, $C_L = 100\mu F$, $T_A = 25^{\circ}C$		1.8		ms
t_F	V_{OUT} Fall Time ^(a,b)			34		ms
t_{OFF}	Turn-Off Time ^(a,d)			35		ms

Notes:

- This parameter is guaranteed by design and characterization; not production tested.
- t_{DON} / t_{DOFF} / t_R / t_F are defined in Figure 2.
- $t_{ON} = t_R + t_{DON}$
- $t_{OFF} = t_F + t_{DOFF}$

ET3158A

Timing Diagram

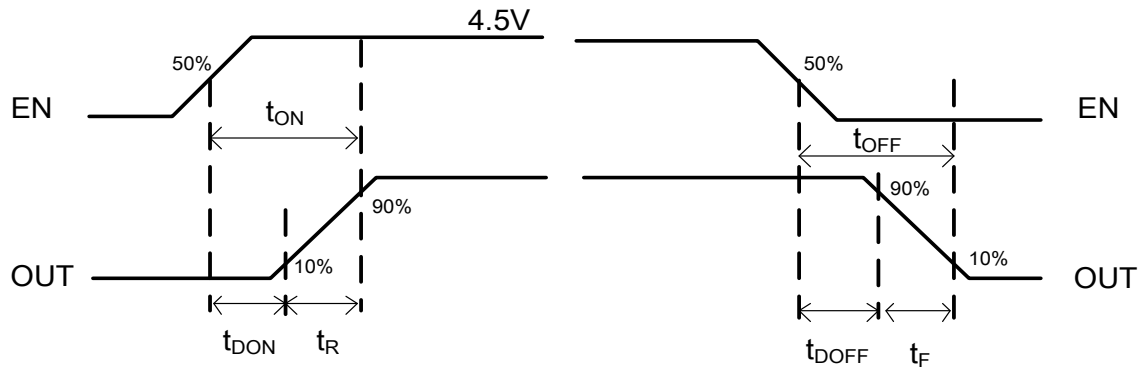


Figure 2. Timing Diagram

Typical Characteristics

Internally regulated $T_A = 25^\circ\text{C}$, unless otherwise noted

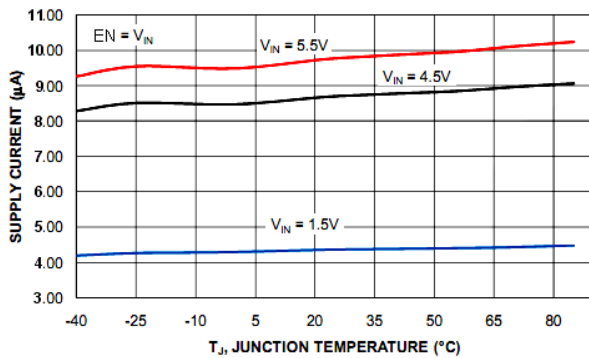


Figure 3. Supply Current vs Supply Voltage

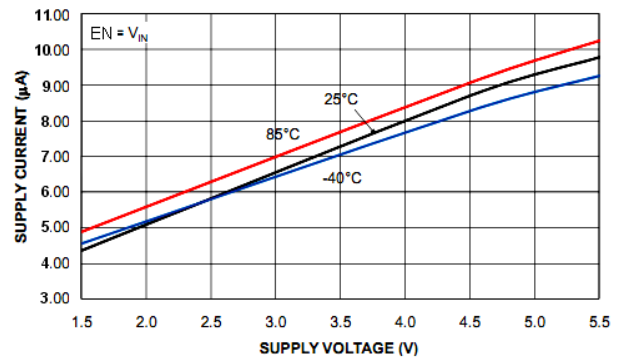


Figure 4. Supply Current vs Temperature

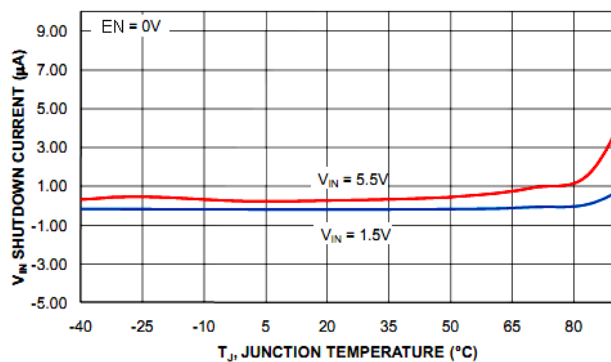


Figure 5. Shutdown Current vs Supply Voltage

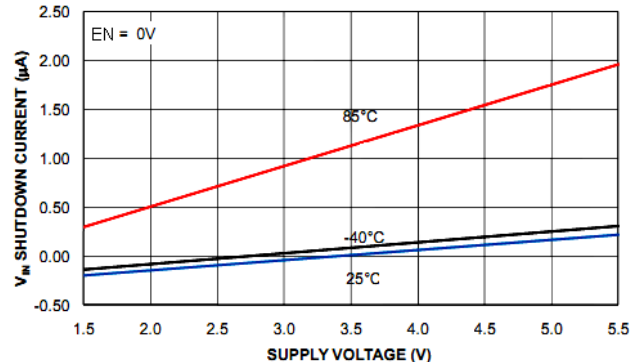


Figure 6. Shutdown Current vs Temperature

ET3158A

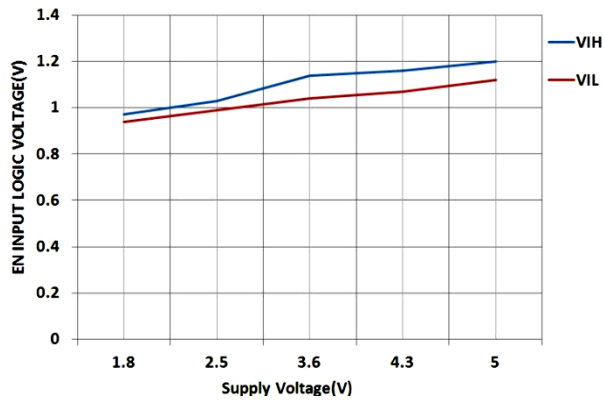


Figure 7. EN Pin Threshold vs Supply Voltage

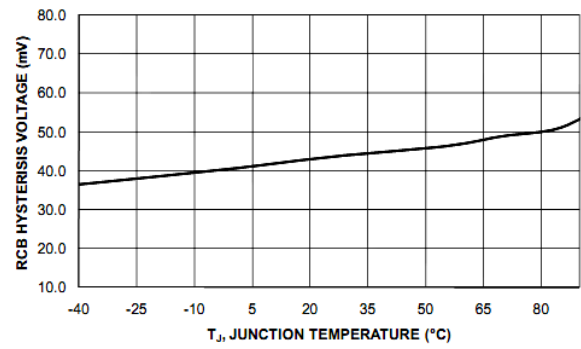


Figure 8. RCB Hysteresis vs Temperature

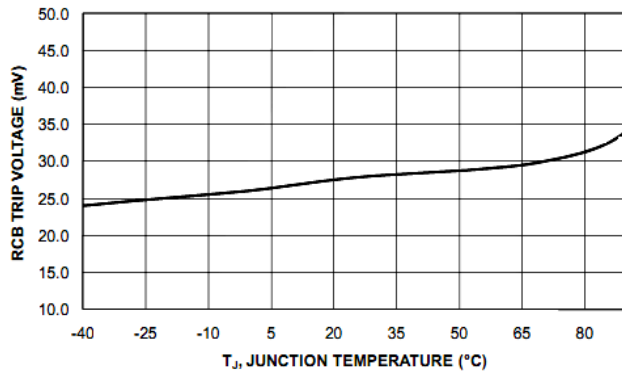


Figure 9. RCB Trip vs Temperature

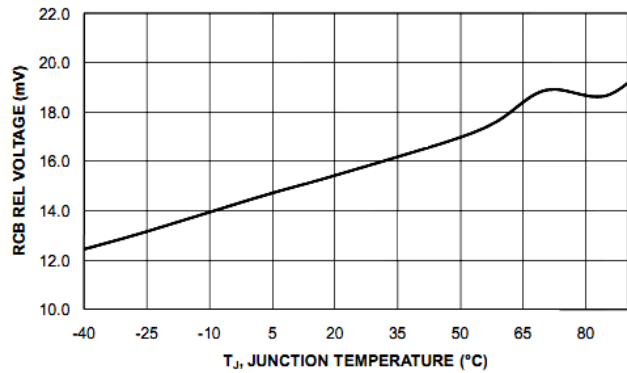


Figure 10. RCB Release vs Temperature

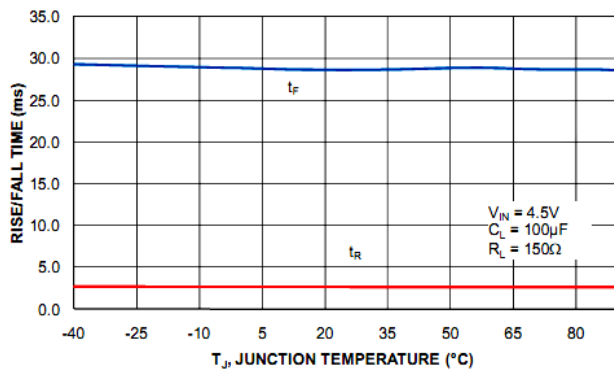


Figure 11. t_R / t_F vs Temperature

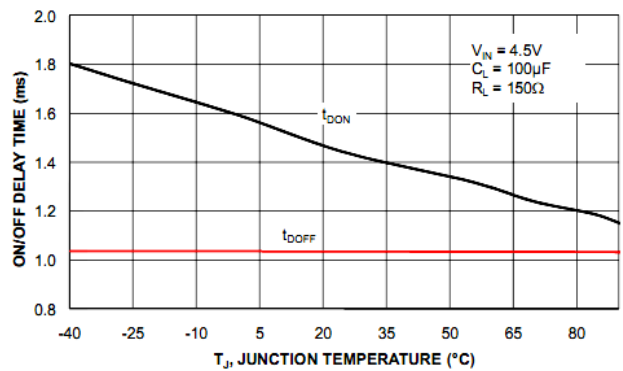
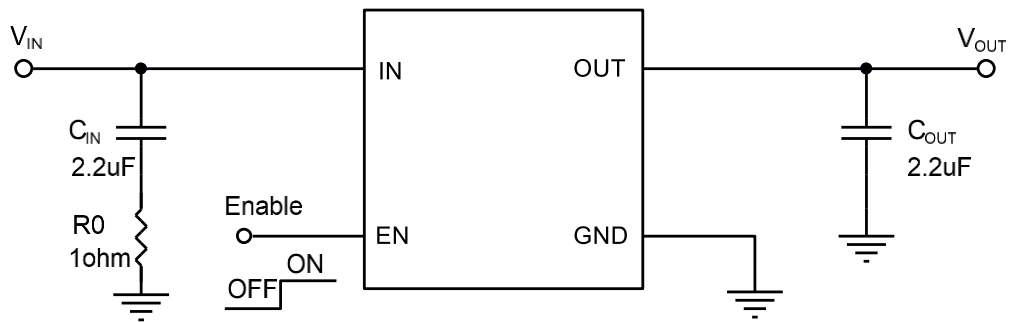


Figure 12. t_{DON} / t_{DOFF} vs Temperature

ET3158A

Application Circuits



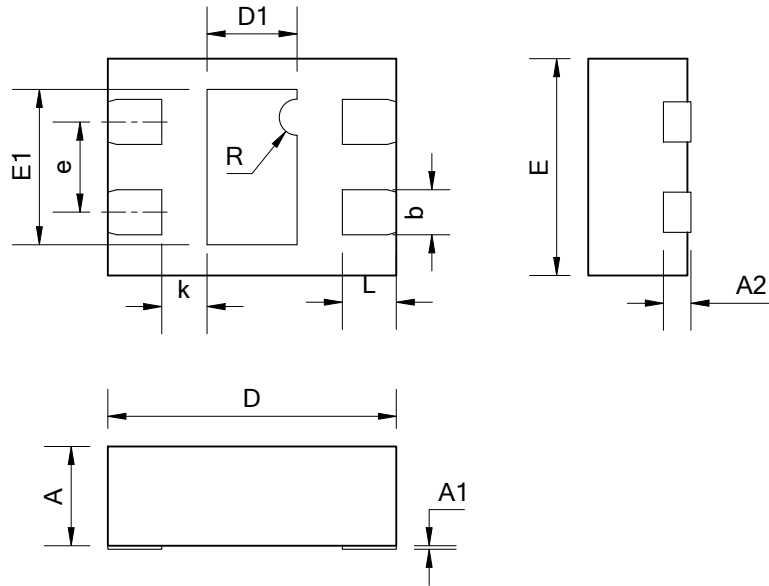
Notes:

- This circuit only supplies for reference;
- Recommended Capacitor C_{IN} value is 2.2 μ F or 1 μ F;
- Recommended Capacitor C_{OUT} value is 4.7 μ F or 2.2 μ F or 1 μ F;
- Recommended R_0 resistor value is 2.2 Ω or 1 Ω .

ET3158A

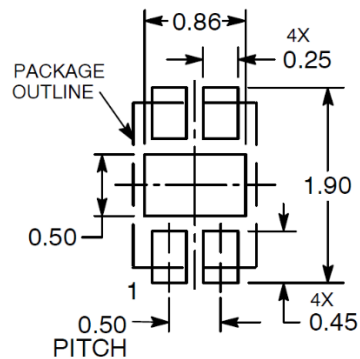
Package Dimension

DFN4



Unit: mm

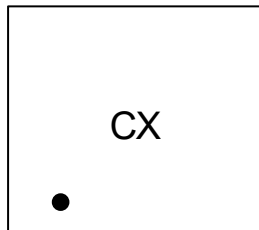
Symbol	Min	Typ	Max	Symbol	Min	Typ	Max
A	0.5	0.55	0.6	D1	0.45	0.5	0.55
A1	0	0.02	0.05	E1	0.81	0.86	0.91
A2	0.152REF			L	0.25	0.3	0.35
D	1.5	1.6	1.7	b	0.2	0.25	0.3
E	1.10	1.20	1.30	e	0.500BSC		
R	0.05	0.1	0.15	k	0.15	0.25	0.35



Recommended Mounting Footprint Unit:mm

ET3158A

Marking Information



C = Part Number
X = Tracking Number

Revision History and Checking Table

Version	Date	Revision Item	Modifier	Function & Spec Checking	Package & Tape Checking
1.2	2023-3-29	Update Typeset	Shi Bo	Liuxm	Zhu Jun Li
1.3	2025-7-23	Update R _{ON}	Wangar	Yangxx	Liuju
1.3.a	2025-12-24	Add Marking Information	Yangxx	Yangxx	Liuju